We extend the result of (Lee & Choi, 2025) to the goal-conditioned, EBM-guided planning setting.

Definition 4.1. *Let $\nabla_{z_t} E_{true}(z_t|c)$ denote the true optimal energy guidance and $\nabla_{z_t} E_\phi(z_t|c)$ be our learned EBM guidance. The guidance gap at z_t is:*

$$\Delta_{EBM}(z_t) = \|\nabla_{z_t} E_{true}(z_t|c) - \nabla_{z_t} E_\phi(z_t|c)\|_2 \tag{9}$$

Proposition 4.2. *(Proof in Appendix) The EBM guidance gap $\Delta_{EBM}(z_t)$ has a lower bound scaling as $\frac{c}{\sqrt{1-\bar{\alpha}_t}}\sqrt{d}$ in high-dimensional latent spaces, where $c > 0$ is a constant independent of dimensionality d.*

Under our contrastive training, successful latent subgoal sequences concentrate on a k-dimensional submanifold $\mathcal{M}_0 \subset \mathbb{R}^d$ with $k \ll d$. To mitigate this manifold deviation, we enhance the standard EBM guidance with a two-step manifold-aware process as suggested in (Lee & Choi, 2025). Rather than applying guidance directly to the noise prediction, we perform:

Step 1: Guided Sampling.

$$z_{\ell-1}^{\text{temp}} \sim \mathcal{N}\big(\mu_\theta(z_\ell) + w_{ebm}\,\Sigma^\ell\, g, \Sigma^\ell\big) \tag{10}$$

where $g = \nabla_{z_\ell} E_\phi(z_\ell|c)$ is the EBM guidance and μ_θ, Σ^ℓ are the mean and covariance of the reverse diffusion transition.

Step 2: Manifold Projection.

$$z_{\ell-1} = \mathcal{P}_{\mathcal{T}_{z_{\ell-1}}\mathcal{M}_{\ell-1}}\big(z_{\ell-1}^{\text{temp}}\big) \tag{11}$$

where $\mathcal{P}_{\mathcal{T}_{z_{\ell-1}}\mathcal{M}_{\ell-1}}$ projects onto the local tangent space of the latent manifold $\mathcal{M}_{\ell-1}$.

The manifold projection is computed using local low-rank approximation: we first obtain a denoised estimate using Tweedie's formula (Robbins, 1992; Chung et al., 2022; 2023):

$$\hat{z}^{0|\ell-1} = \frac{1}{\sqrt{\bar{\alpha}_{\ell-1}}}\Big(z_{\ell-1}^{\text{temp}} - \sqrt{1-\bar{\alpha}_{\ell-1}}\,\epsilon_\theta(z_{\ell-1}^{\text{temp}}, \ell-1, c)\Big) \tag{12}$$

We then retrieve k nearest neighbors from successful latent subgoal sequences using cosine similarity (Feng et al., 2024), forward diffuse them to timestep $\ell - 1$, and perform rank-r PCA to obtain the projection basis $\boldsymbol{U} \in \mathbb{R}^{d\times r}$. Let $\boldsymbol{\mu}$ be the local mean of these neighbors. The mean-centered projection is

$$\mathcal{P}(\mathbf{z}) = \boldsymbol{\mu} + \boldsymbol{U}\boldsymbol{U}^T(\mathbf{z} - \boldsymbol{\mu}).$$

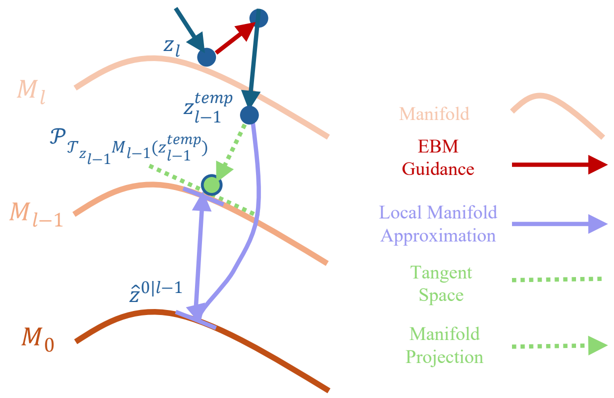

Figure 2. **Manifold-Aware EBM-Guided Diffusion step**: Starting from a noisy latent sample ($\boldsymbol{z}_\ell$), the standard reverse diffusion predicts a mean. EBM guidance (red arrow) then shifts this mean, leading to a guided sample ($\boldsymbol{z}_{\ell-1}^{\text{temp}}$). To prevent manifold deviation, $\boldsymbol{z}_{\ell-1}^{\text{temp}}$ is subsequently projected onto the local latent manifold $\mathcal{M}_{\ell-1}$ (using local manifold approximation and projection, indicated by purple and green arrows), yielding the final projected sample ($\boldsymbol{z}_{\ell-1}$). This two-step process ensures both successful and feasible subgoal generation within the hierarchical planning framework.

Proposition 4.3. *Given a base diffusion planner ϵ_θ trained for classifier-free guidance, a learned energy function $E_\phi(z|c)$, and a manifold projection operator $\mathcal{P}_\mathcal{M}$, the manifold-aware guided planner, which combines EBM guidance with manifold projection, corresponds to sampling from a posterior distribution $p(z|y = 1, z \in \mathcal{M}, c)$ that maximizes the likelihood of generating a successful and feasible goal-conditioned plan.*

4.2.2. Low-Level Planner: Rectified Flow for Trajectory Generation

The low-level planner's role is to generate a dense, short-horizon latent trajectory τ_z to a given subgoal z_k. We can frame this subproblem through the lens of optimal transport. Here, the task is to find the optimal mapping from the distribution of initial states around z_{k-1} to the distribution of target states around z_k. The *cost* of transport is minimized by trajectories that are as straight as possible in the latent space. We use a conditional rectified flow model, $v_\theta(\tau_u, u, c_k)$, for its speed. It is trained to generate a trajectory segment τ conditioned on the context $c_k = (z_{k-1}, z_k)$ by minimizing the standard flow-matching objective from Eq. 2:

$$\mathcal{L}_{LL} = \mathbb{E}_{u,\tau_0,\tau_1}\Big[\big\|v_\theta\big((1-u)\tau_0 + u\tau_1, u, c_k\big) - (\tau_1 - \tau_0)\big\|^2\Big] \tag{13}$$

Construction of training pairs. For each consecutive latent subgoal pair (z_{k-1}, z_k), we extract the dense H-step latent